EE3011.ELECTRONICS 1

Experiment (6)

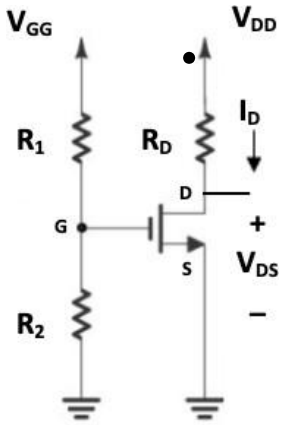
Mosfet IV Characteristics

Mehmet Ali Kamalak

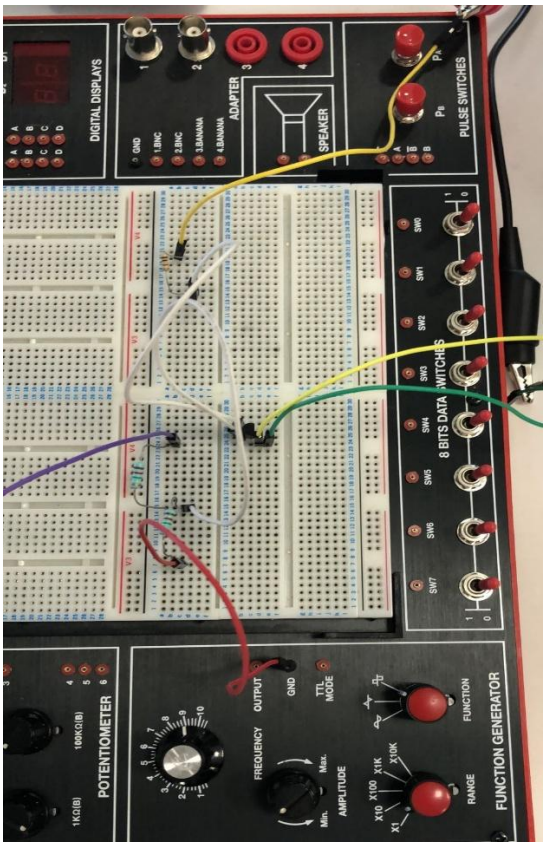
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Part (1): MOSFET IV Characteristics

For the first part, I connected to circuit as it is and give the values of 1Mega Ohm to R1 and R2, 1k to RD and +8V to VGG. Then I change the voltage of VDD as starting from 0 to 10V.



This is the circuit that I have.

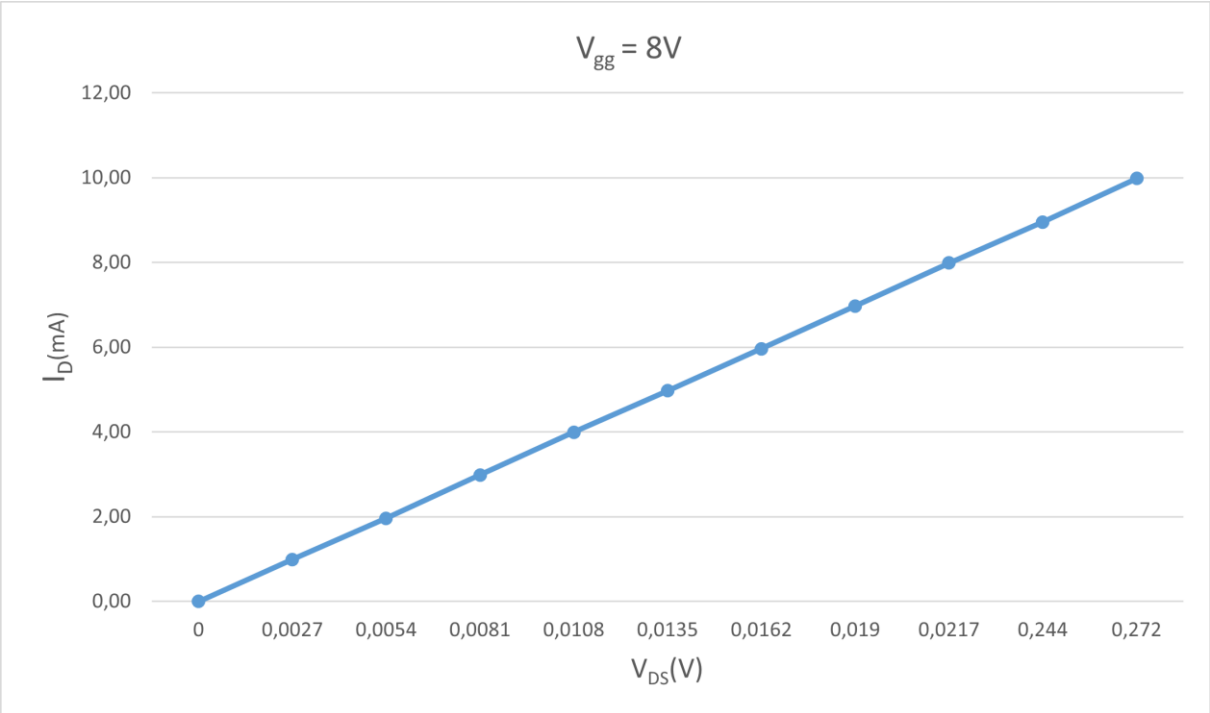


Because there is no 1 MegaΩ resistor I connected 10 10kΩ resistors to get 1 MegaΩ. We use this much resistor to ignore the value. Then I record the voltage and current values and merge them in table 2 and graph 1.

Vdd(V)	Vds	Vrd	Id	Vds	Vrd	Id
	0	0	0	0	0	0
1	5.4mV	994.6 mV	994.6 µA	3.096mV	996.8 mV	996.8 µA
2	10.9 mV	1.988V	1.988mA	6.197 mV	1.993 V	1.993 mA
3	16.4 mV	2.983V	2.983 mA	9.304 mV	2.990 V	2.990 mA
4	21.9 mV	3.977V	3.977 mA	12.414mV	3.987 V	3.987 mA
5	27.5 mV	4.972V	4.972 mA	15.531mV	4.984 V	4.984 mA
6	33.2 mV	5.967V	5.967 mA	18.652mV	5.981 V	5.981 mA
7	38.8 mV	6.961V	6.961 mA	21.980mV	6.978 V	6.978 mA
8	44.5 mV	7.954V	7.954 mA	24.915mV	7.975 V	7.975 mA
9	50.3 mV	8.948V	8.948 mA	28.050mV	8.972 V	8.972 mA
10	56.1 mV	9.942V	9.942 mA	31.198mV	9.969 V	9.969 mA

Table 2. IV Characteristics Measurements

Graph 1. Mosfet IV Curves



1. Comment on the IV curves that you obtained. Can you identify the transistor modes of operation?

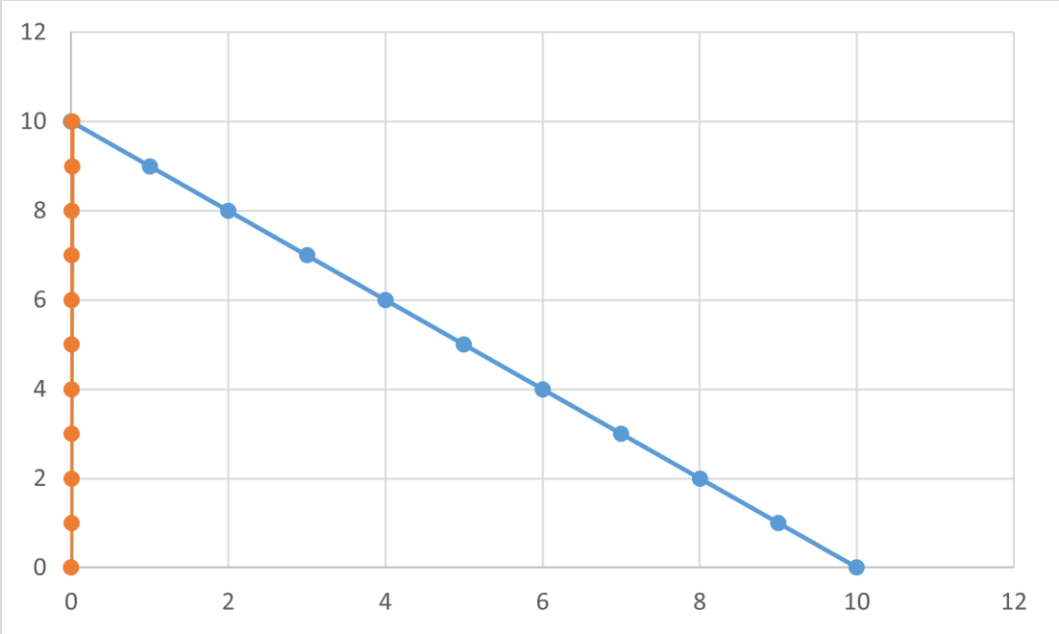
Answer: The mosfet is in saturation mode and the IV curve is linearly increasing.

2. Is the drain current constant in the saturation region of the IV curve? Explain why or why not.

Answer: Drain current is not constant it changes according to r0.

Part (2): Graphical Analysis of a DC MOSFET Circuit.

We asked to draw the load line on top of the IV curves in graph 1. I marked the intersection point of the curves.



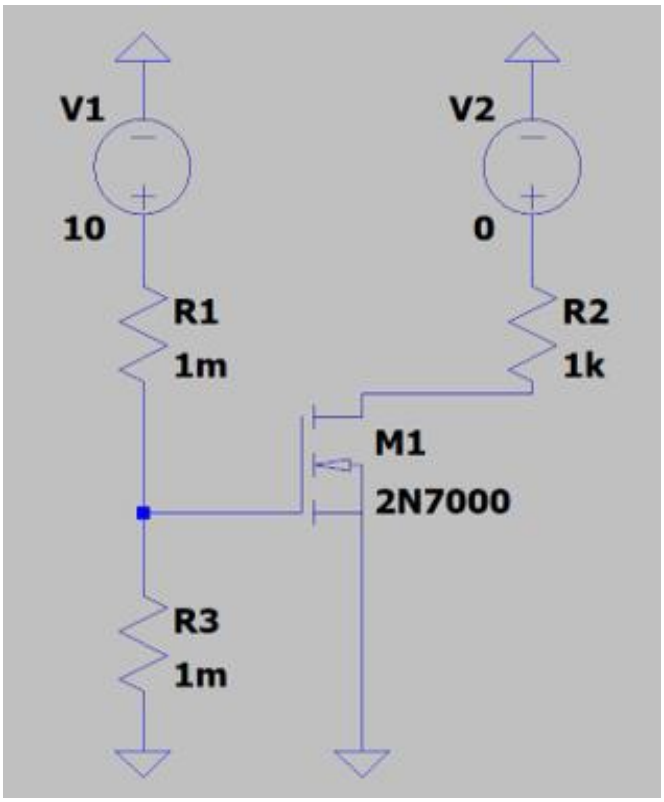
The experimental value of $I_D = (V_{DD} - V_{DS})/R_D$.

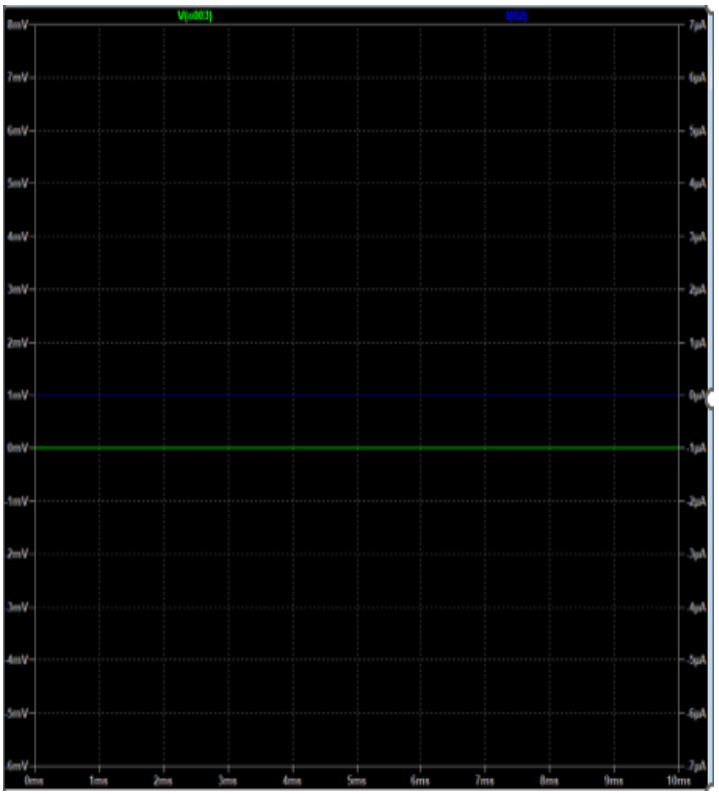
Then I record the results in table 3.

Table 3. Graphical Solution Results.

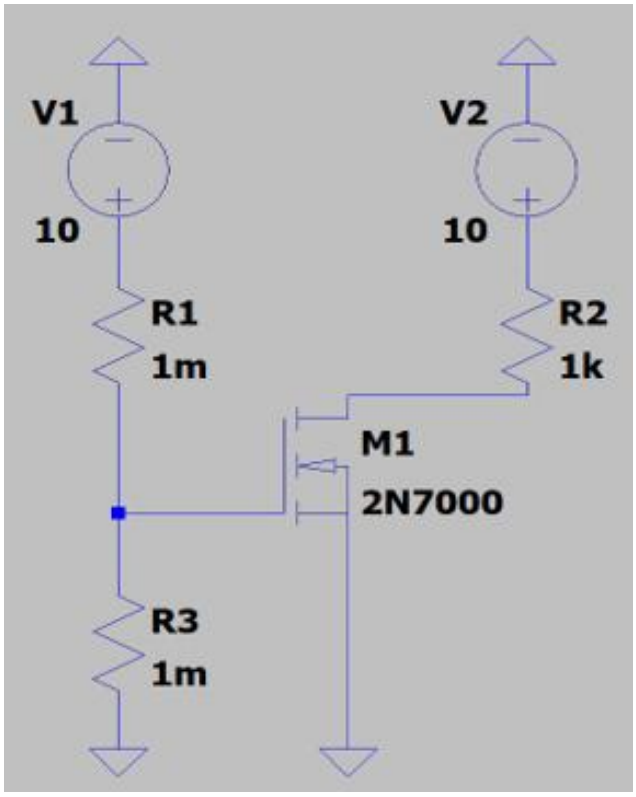
V_{DD}= 10 V and V_{GG} = 10 V					
V_{DS} (V) Experimental value	V_{DS} (V) Graphical value	% Difference	I_D (mA) Experimental value	I_D (mA) Graphical value	% Difference
21.9mV	24.6	%12.33	3.977mA	4.456mA	%12.09

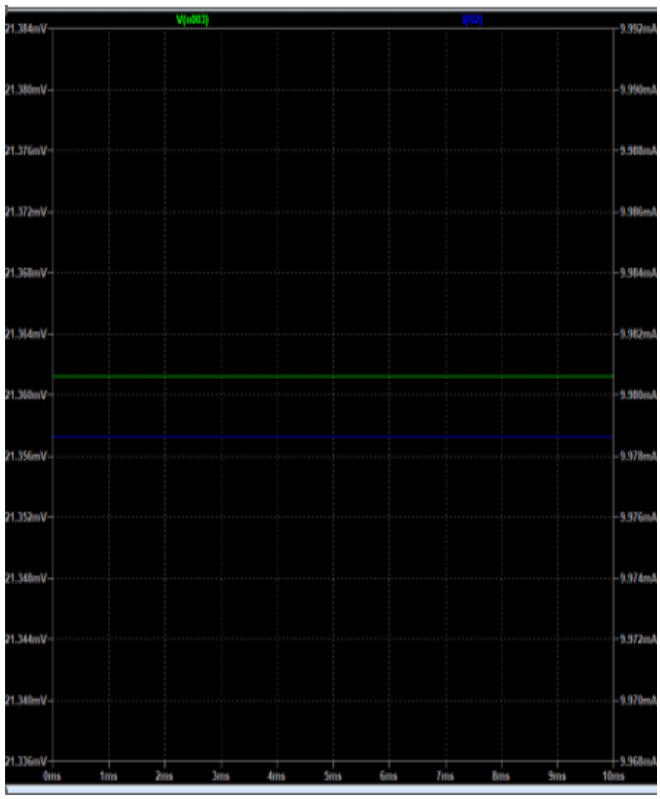
If we increase the voltage value more slowly, the error decreases.



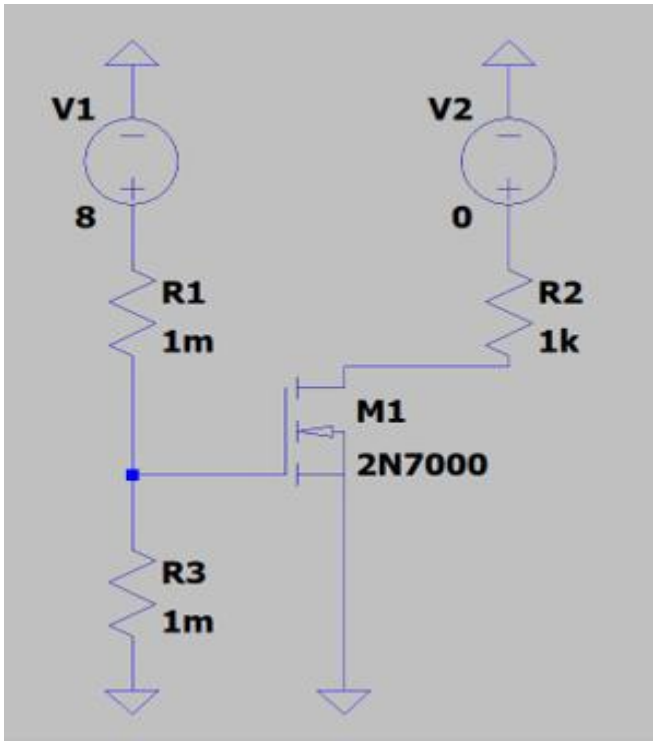


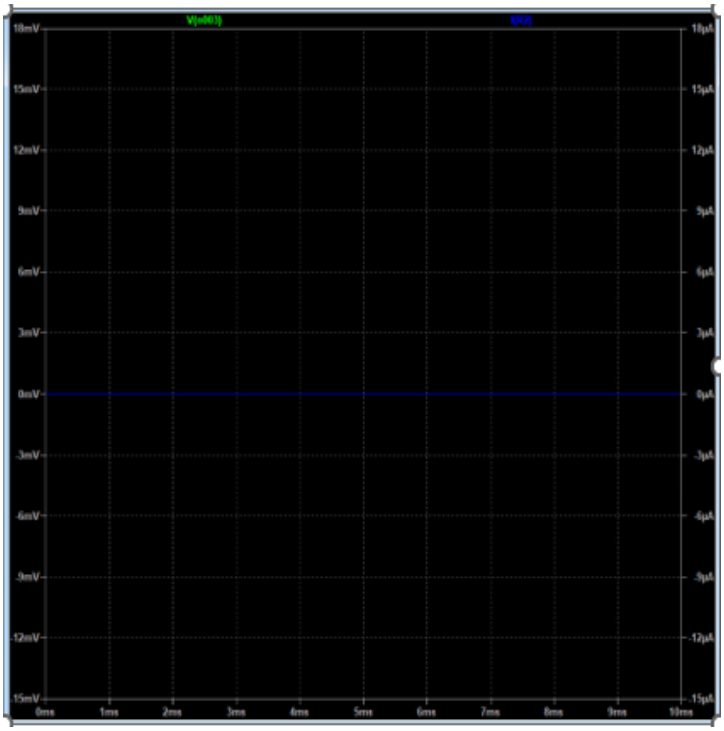
V_{DD}= 0 V and V_{GG} = 10 V



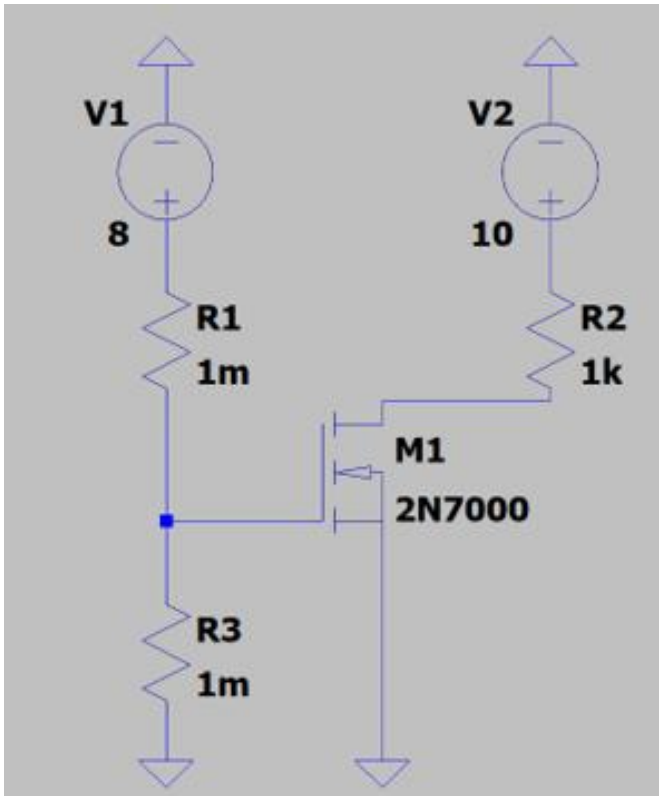


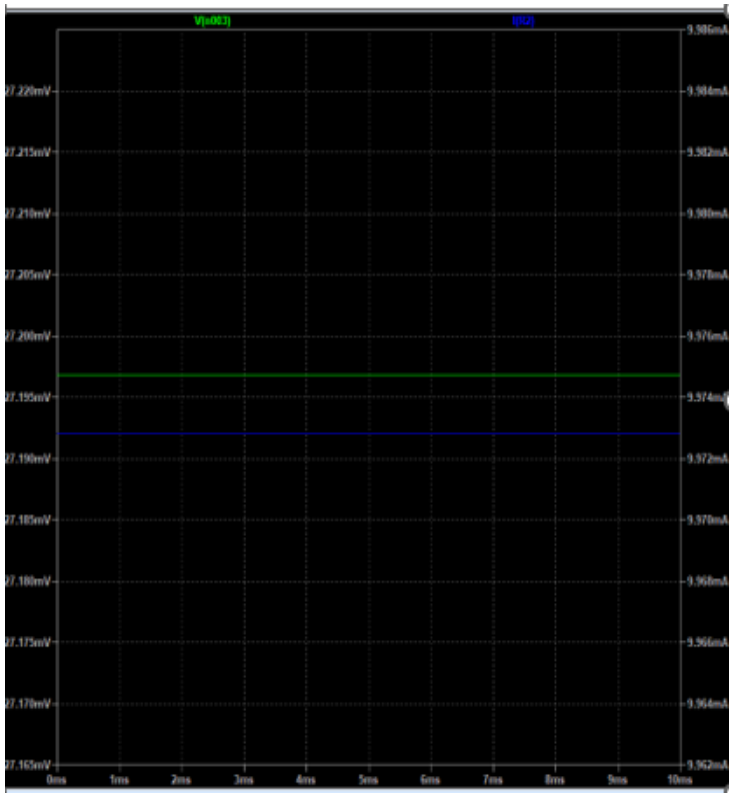
V_{DD}= 10 V and V_{GG} = 10 V





V_{DD}= 0 V and V_{GG} = 8 V





V_{DD}= 10 V and V_{GG} = 8 V

These are the Ltspice simulations. I don't need to put all the experimental values. I am sure of my measurements and calculations. I hope it is correct.